Lectures 12–15 Martingales

Conditional expectation · stopping and convergence · the martingale CLT · concentration

Reading guide. Section 1 is a reminder of L^2 and orthogonal projection, and defines conditional expectation. Sections 1–2 are Lecture 12; Section 3 is Lecture 13; Section 4 is Lecture 14; Section 5 is Lecture 15. Everything is done in L^2. Facts taken from outside the course are in boxes marked QUOTED. Examples marked WORKED IN CLASS are done on the board. Exercises are typed A (routine), B (a worked example or short proof) and C (longer; AI assistance encouraged); the homework for a lecture is the exercise block closing its section.

1 Reminder: L^2, projection, conditional expectation

1.1 L^2 and orthogonal projection

L^2 is the space of real random variables X with $\mathbb{E}X^2 < \infty$, with inner product and norm

$$\langle X, Z\rangle = \mathbb{E}[XZ], \qquad \|X\|_2 = \big(\mathbb{E}X^2\big)^{1/2}.$$

For a finite family of random variables $Y = (Y_1, \dots, Y_n)$ (with values in any $\mathbb{R}^m$) put

$$L^2(Y) = \big\{\, f(Y) \;:\; f : \mathbb{R}^m \to \mathbb{R},\ \ \mathbb{E}f(Y)^2 < \infty \,\big\},$$

the functions of Y with finite second moment. It is a linear subspace of L^2 containing the constants, and $L^2(Y) \subset L^2(Y, Y')$ for any further family Y'.

QUOTED · Hilbert space

L^2 is complete: a sequence that is Cauchy in $\|\cdot\|_2$ converges in $\|\cdot\|_2$. Each $L^2(Y)$ is a closed subspace. For a closed subspace $V \subset L^2$ and $X \in L^2$ there is a unique $P_V X \in V$ with $\|X - P_V X\|_2 = \min_{W \in V} \|X - W\|_2$, characterized by $\langle X - P_V X, W\rangle = 0$ for all $W \in V$. The map P_V is linear with $\|P_V X\|_2 \le \|X\|_2$; $\|X\|_2^2 = \|P_V X\|_2^2 + \|X - P_V X\|_2^2$; and if $V \subset W$ are closed subspaces then $P_V P_W = P_W P_V = P_V$.

1.2 Conditional expectation

Definition 1.1 (conditional expectation)**.** For $X \in L^2$ and a finite family Y, $\mathbb{E}[X \mid Y] := P_{L^2(Y)} X$ is the orthogonal projection of X onto $L^2(Y)$.

This is the definition of Lectures 1–2: $\mathbb{E}[X \mid Y]$ is the minimizer of $\mathbb{E}(X - f(Y))^2$ over all functions f with $f(Y) \in L^2$, i.e. the best mean-square prediction of X by any function of Y. The best *linear* prediction is the projection onto the smaller subspace $\operatorname{span}(1, Y_1, \dots, Y_n)$; the two differ in general (Example 1.7) and coincide for Gaussian vectors (Theorem 1.6).

Lemma 1.2 (characterization)**.** *Let $Z \in L^2(Y)$. Then $Z = \mathbb{E}[X \mid Y]$ if and only if $\mathbb{E}[(X - Z)f(Y)] = 0$ for every $f(Y) \in L^2(Y)$, and it suffices to check bounded f.*

Proof. The first statement is the characterization in the box. For the second, truncate: $f_M = \max(-M, \min(f, M))$ satisfies $f_M(Y) \to f(Y)$ in L^2, and $\mathbb{E}[(X - Z)f_M(Y)] \to \mathbb{E}[(X - Z)f(Y)]$ by Cauchy–Schwarz. □

Proposition 1.3 (rules)**.** *Let $X, X' \in L^2$, $a, b \in \mathbb{R}$, and let Y, Y' be finite families.*

(i) Linearity. $\mathbb{E}[aX + bX' \mid Y] = a\mathbb{E}[X \mid Y] + b\mathbb{E}[X' \mid Y]$.

(ii) Averaging. $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}X$.

(iii) Taking out what is known. *If $g(Y) \in L^2(Y)$ and $g(Y)X \in L^2$, then $g(Y)\mathbb{E}[X \mid Y] \in L^2$ and $\mathbb{E}[g(Y)X \mid Y] = g(Y)\mathbb{E}[X \mid Y]$; in particular $\mathbb{E}[g(Y) \mid Y] = g(Y)$.*

(iv) Tower. $\mathbb{E}\big[\mathbb{E}[X \mid Y, Y'] \,\big|\, Y\big] = \mathbb{E}[X \mid Y] = \mathbb{E}\big[\mathbb{E}[X \mid Y] \,\big|\, Y, Y'\big]$.

(v) Independence. *If X is independent of Y then $\mathbb{E}[X \mid Y] = \mathbb{E}X$.*

(vi) Positivity. *If $X \geq 0$ then $\mathbb{E}[X \mid Y] \geq 0$; hence $X \leq X'$ implies $\mathbb{E}[X \mid Y] \leq \mathbb{E}[X' \mid Y]$, and $|\mathbb{E}[X \mid Y]| \leq \mathbb{E}[|X| \mid Y]$.*

(vii) Pythagoras. $\mathbb{E}X^2 = \mathbb{E}\big(\mathbb{E}[X \mid Y]\big)^2 + \mathbb{E}\big(X - \mathbb{E}[X \mid Y]\big)^2$.

(viii) Contraction. $\mathbb{E}\big(\mathbb{E}[X \mid Y]\big)^2 \leq \mathbb{E}X^2$, *and* $\mathbb{E}\big(\mathbb{E}[X \mid Y] - \mathbb{E}[X' \mid Y]\big)^2 \leq \mathbb{E}(X - X')^2$.

Proof. (i), (iv), (vii), (viii) are properties of orthogonal projections listed in the box, (iv) because $L^2(Y) \subset L^2(Y, Y')$. Write $Z = \mathbb{E}[X \mid Y]$ and use Lemma 1.2. (ii) Test against $f \equiv 1$. (iii) For bounded g: $g(Y)Z \in L^2(Y)$ and $\mathbb{E}[(g(Y)X - g(Y)Z)f(Y)] = \mathbb{E}[(X - Z)(gf)(Y)] = 0$ for bounded f. For general g truncate, $g_M = \max(-M, \min(g, M))$. Then $g_M(Y)X \to g(Y)X$ in L^2, so $\mathbb{E}[g_M(Y)X \mid Y] \to \mathbb{E}[g(Y)X \mid Y]$ in L^2 by (viii); and $\mathbb{E}[g_M(Y)X \mid Y] = g_M(Y)Z$ with $\|g_M(Y)Z\|_2 \leq \|g_M(Y)X\|_2 \leq \|g(Y)X\|_2$ for every M, so letting $M \to \infty$, $g(Y)Z \in L^2$ and $g_M(Y)Z \to g(Y)Z$ in L^2. The two limits agree. (v) $\mathbb{E}[(X - \mathbb{E}X)f(Y)] = \mathbb{E}[X - \mathbb{E}X]\,\mathbb{E}f(Y) = 0$. (vi) Test against $f(Y) = \mathbf{1}\{Z < 0\}$: $\mathbb{E}[Z\mathbf{1}\{Z < 0\}] = \mathbb{E}[X\mathbf{1}\{Z < 0\}] \geq 0$, so $\mathbb{P}(Z < 0) = 0$; the rest follows by linearity applied to $X' - X \geq 0$ and to $\pm X \leq |X|$. □

Definition 1.4 (conditional variance)**.** $\operatorname{Var}(X \mid Y) := \mathbb{E}\big[(X - \mathbb{E}[X \mid Y])^2 \,\big|\, Y\big]$. By (ii) and (vii), $\operatorname{Var} X = \mathbb{E}\operatorname{Var}(X \mid Y) + \operatorname{Var}\mathbb{E}[X \mid Y]$.

Lemma 1.5 (freezing)**.** *Let $U \in L^2(Y)$ take values in $\mathbb{R}^p$, let Z be a random variable in $\mathbb{R}^q$ independent of Y, and let $h : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}$ satisfy $h(U, Z) \in L^2$. Put $\bar{h}(u) = \mathbb{E}\,h(u, Z)$. Then*

$$\mathbb{E}\big[h(U, Z) \mid Y\big] = \bar{h}(U).$$

Proof. $\bar{h}(U) \in L^2(Y)$. For bounded f, independence of Z from (U, Y) allows integrating over Z first: $\mathbb{E}[h(U, Z)f(Y)] = \mathbb{E}\big[\bar{h}(U)f(Y)\big]$. Apply Lemma 1.2. □

Theorem 1.6 (Gaussian conditioning, Lectures 1–2)**.** *Let (X, Y) be jointly Gaussian in $\mathbb{R}^p \times \mathbb{R}^m$ with Σ_{YY} invertible, and put $A = \Sigma_{XY}\Sigma_{YY}^{-1}$. Then $X - AY$ is independent of Y, and*

$$\mathbb{E}[X \mid Y] = \mu_X + A(Y - \mu_Y), \qquad \operatorname{Var}(X \mid Y) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}$$

(componentwise in the first identity; the second is the conditional covariance matrix, which does not depend on Y). That is, for jointly Gaussian (X, Y) the best prediction among all functions of Y is the linear one, and the conditional law of X given Y is Gaussian with the displayed mean and covariance.

Example 1.7 (a non-Gaussian pair)**.** Let Y be uniform on $[-1, 1]$ and $X = Y^2$ (the failure case of Lecture 2). Then $\operatorname{Cov}(X, Y) = \mathbb{E}Y^3 = 0$, so the best linear prediction of X from Y is the constant $\mathbb{E}X = \frac{1}{3}$, with residual variance $\operatorname{Var} Y^2 = \frac{4}{45}$; while $\mathbb{E}[X \mid Y] = Y^2$ by rule (iii), with residual zero.

2 Martingales

2.1 Filtrations

Fix a sequence of random variables $Y_1, Y_2, \dots$ (each with values in some $\mathbb{R}^m$). For $n \geq 0$ we define

$$\mathbb{E}[X \mid \mathcal{F}_n] := \mathbb{E}[X \mid Y_1, \dots, Y_n], \qquad L^2(\mathcal{F}_n) := L^2(Y_1, \dots, Y_n),$$

with $\mathbb{E}[X \mid \mathcal{F}_0] := \mathbb{E}X$ and $L^2(\mathcal{F}_0)$ the constants. We abbreviate $\mathbb{E}_n[X] := \mathbb{E}[X \mid \mathcal{F}_n]$. The symbol $\mathcal{F}_n$ stands for "the first n variables"; the sequence $(\mathcal{F}_n)_{n\geq 0}$ is the *filtration* generated by (Y_n). By the tower rule,

$$\mathbb{E}_m \mathbb{E}_n = \mathbb{E}_n \mathbb{E}_m = \mathbb{E}_m \qquad (m \leq n).$$

Definition 2.1. A sequence $(X_n)_{n\geq 0}$ is *adapted* if $X_n \in L^2(\mathcal{F}_n)$ for every n, and *predictable* if $X_n \in L^2(\mathcal{F}_{n-1})$ for every $n \geq 1$.

2.2 Martingales and orthogonal increments

Definition 2.2 (martingale)**.** Let $(\mathcal{F}_n)$ be the filtration generated by a sequence (Y_n). A sequence $(M_n)_{n\geq 0}$ in L^2 is a *martingale* with respect to $(\mathcal{F}_n)$ if it is adapted and

$$\mathbb{E}_n[M_{n+1}] = M_n \qquad \text{for all } n \geq 0;$$

a *submartingale* if it is adapted and $\mathbb{E}_n[M_{n+1}] \geq M_n$; a *supermartingale* if it is adapted and $\mathbb{E}_n[M_{n+1}] \leq M_n$. The *differences* of M are $\xi_n = M_n - M_{n-1}$, $n \geq 1$; the martingale property reads $\mathbb{E}_{n-1}[\xi_n] = 0$.

The filtration is part of the data. Every sequence (M_n) generates its own filtration (take $Y_n = M_n$), and a martingale with respect to a filtration $(\mathcal{F}_n)$ is also a martingale with respect to its own (adaptedness gives $L^2(M_0, \dots, M_n) \subset L^2(\mathcal{F}_n)$; apply the tower rule), but not conversely. Below, "martingale" means martingale with respect to the filtration of Section 2.1 unless stated otherwise.

By the tower rule, $\mathbb{E}_m[M_n] = M_m$ for $m \leq n$, and $\mathbb{E}M_n = \mathbb{E}M_0$. Conversely, if $(\xi_n)_{n\geq 1}$ is adapted with $\mathbb{E}_{n-1}\xi_n = 0$, then $M_n = M_0 + \sum_{k\leq n} \xi_k$ is a martingale for any constant M_0.

Lemma 2.3 (orthogonal increments)**.** *Let M be a martingale with differences ξ_k. If $W \in L^2(\mathcal{F}_{n-1})$ then $\mathbb{E}[W\xi_n] = 0$. Consequently $\mathbb{E}[\xi_m \xi_n] = 0$ for $m \neq n$, and*

$$\mathbb{E}M_n^2 = \mathbb{E}M_0^2 + \sum_{k=1}^{n} \mathbb{E}\xi_k^2, \qquad \mathbb{E}(M_n - M_m)^2 = \sum_{k=m+1}^{n} \mathbb{E}\xi_k^2 \quad (m \leq n).$$

Proof. $\mathbb{E}_{n-1}\xi_n = 0$ says that the projection of ξ_n onto $L^2(\mathcal{F}_{n-1})$ is 0, i.e. $\xi_n \perp L^2(\mathcal{F}_{n-1})$ (Lemma 1.2 with $Z = 0$). Since M_0 and ξ_m, $m < n$, lie in $L^2(\mathcal{F}_{n-1})$, the terms of $M_n = M_0 + \xi_1 + \cdots + \xi_n$ are pairwise orthogonal, and the identities are Pythagoras. □

2.3 Examples

Example 2.4. (a) *Sums of independent variables.* WORKED IN CLASS Let $Y_1, Y_2, \dots$ be independent and centered with variances $s_k^2 = \mathbb{E}Y_k^2$, and let $S_n = \sum_{k\leq n} Y_k$. Then

$$S_n, \qquad S_n^2 - \sum_{k\leq n} s_k^2, \qquad \frac{e^{\lambda S_n}}{\prod_{k\leq n} \mathbb{E}e^{\lambda Y_k}} \quad (\lambda \in \mathbb{R},\ \mathbb{E}e^{2\lambda Y_k} < \infty)$$

are martingales. Indeed, by rules (iii) and (v), $\mathbb{E}_n S_{n+1} = S_n + \mathbb{E}Y_{n+1} = S_n$; $\mathbb{E}_n[S_{n+1}^2] = S_n^2 + 2S_n\mathbb{E}Y_{n+1} + \mathbb{E}Y_{n+1}^2 = S_n^2 + s_{n+1}^2$; and $\mathbb{E}_n[e^{\lambda S_{n+1}}] = e^{\lambda S_n}\,\mathbb{E}e^{\lambda Y_{n+1}}$. For the simple random walk ($Y_k = \pm 1$ with probability $\frac{1}{2}$) these are S_n, $S_n^2 - n$ and $e^{\lambda S_n}/(\cosh\lambda)^n$.

(b) *Exponential martingale, i.i.d. case.* If the Y_k are i.i.d. with $\psi(\lambda) = \log \mathbb{E}e^{\lambda Y_1}$, the third martingale in (a) is $M_n = \exp(\lambda S_n - n\psi(\lambda))$.

(c) *Doob martingale.* For $f(Y_1, \dots, Y_N) \in L^2$, the sequence $M_n = \mathbb{E}_n[f(Y_1, \dots, Y_N)]$, $0 \le n \le N$, is a martingale by the tower rule, with $M_0 = \mathbb{E}f$, $M_N = f$. Hence $f - \mathbb{E}f = \sum_{k=1}^{N} \xi_k$ and, by Lemma 2.3, $\operatorname{Var} f = \sum_{k \le N} \mathbb{E}\xi_k^2$.

(d) *Stochastic gradient descent.* Let $\ell(\theta; z)$ be a loss, $\theta \in \mathbb{R}^d$, with population loss $L(\theta) = \mathbb{E}\,\ell(\theta; z)$, and assume $\mathbb{E}\|\nabla\ell(\theta; z)\|^2 < \infty$ for every θ. Let $z_1, z_2, \dots$ be i.i.d. samples, $\mathcal{F}_n = (z_1, \dots, z_n)$, let $\eta_0, \eta_1, \dots$ be deterministic step sizes, and let

$$\theta_{n+1} = \theta_n - \eta_n \nabla\ell(\theta_n; z_{n+1}), \qquad n \ge 0, \quad \theta_0 \text{ fixed.}$$

Then θ_n is a function of $z_1, \dots, z_n$, and

$$\theta_{n+1} = \theta_0 - \sum_{k=0}^{n} \eta_k \nabla L(\theta_k) - M_{n+1}, \qquad M_{n+1} := \sum_{k=0}^{n} \eta_k \big(\nabla\ell(\theta_k; z_{k+1}) - \nabla L(\theta_k)\big), \quad M_0 := 0.$$

$(M_n)_{n \ge 0}$ is a martingale with respect to $(\mathcal{F}_n)$, coordinatewise: it is adapted, and its differences $\xi_{k+1} = \eta_k(\nabla\ell(\theta_k; z_{k+1}) - \nabla L(\theta_k))$ satisfy $\mathbb{E}_k\xi_{k+1} = 0$ by the freezing lemma with $U = \theta_k \in L^2(\mathcal{F}_k)$ and $Z = z_{k+1}$ independent of $\mathcal{F}_k$, which gives $\mathbb{E}_k[\nabla\ell(\theta_k; z_{k+1})] = \nabla L(\theta_k)$. So the SGD iterate is θ_0, minus the accumulated population-gradient steps, minus a martingale.

Example 2.5 (likelihood ratios)**.** WORKED IN CLASS Let $Y_1, Y_2, \dots$ be i.i.d. with density p, and let q be another density with $\int q^2/p < \infty$. Then

$$M_n = \prod_{k=1}^{n} \frac{q(Y_k)}{p(Y_k)}$$

is a martingale: $\mathbb{E}_n M_{n+1} = M_n\,\mathbb{E}\big[q(Y_{n+1})/p(Y_{n+1})\big] = M_n \int q = M_n$. It is nonnegative with $\mathbb{E}M_n = 1$; $M_n \to 0$ in probability if $q \ne p$ (Exercise 2.10), so a martingale can converge to 0 while its expectation stays 1.

Example 2.6 (Pólya's urn)**.** WORKED IN CLASS An urn holds r red and b black balls. At each step a ball is drawn uniformly at random and returned with one more ball of the same colour. Let R_n be the number of red balls after n draws, $Y_n = \mathbf{1}\{\text{draw } n \text{ is red}\}$, and $F_n = R_n/(r + b + n)$. Then $\mathbb{E}_n[R_{n+1}] = R_n + F_n$, so

$$\mathbb{E}_n[F_{n+1}] = \frac{R_n + F_n}{r + b + n + 1} = \frac{F_n(r + b + n) + F_n}{r + b + n + 1} = F_n :$$

F_n is a martingale with values in $[0, 1]$. Its differences are $\xi_{n+1} = (Y_{n+1} - F_n)/(r + b + n + 1)$, with $\mathbb{E}_n\xi_{n+1}^2 = F_n(1 - F_n)/(r + b + n + 1)^2 \le \frac{1}{4}(r + b + n + 1)^{-2}$, so $\sum_n \mathbb{E}\xi_n^2 < \infty$. Section 3 shows this forces F_n to converge; Exercise 3.16 shows the limit is random.

Exercises for Lecture 12

Exercise 2.7. TYPE A Derive rules (iii) and (iv) of Proposition 1.3 directly from the box in Section 1.1, i.e. from $P_V P_W = P_V$ for $V \subset W$ and from $g(Y)Z \in L^2(Y)$ when $Z \in L^2(Y)$, g bounded.

Exercise 2.8. type A Let (X, Y) be jointly Gaussian, X scalar, with Σ_{YY} invertible. Using Theorem 1.6 and the rules, compute $\mathbb{E}[X^2 \mid Y]$ and $\mathbb{E}[e^X \mid Y]$.

Exercise 2.9. type A Let $Y_1, \dots, Y_N$ be independent with means μ_k. Write out the Doob martingale of $f = \sum_{k \le N} Y_k$ and of $f = \prod_{k \le N} Y_k$, and its differences in each case.

Exercise 2.10. type B In Example 2.5, show $\mathbb{E}\sqrt{M_n} = \rho^n$ with $\rho = \int \sqrt{pq} < 1$ when $q \neq p$, and deduce $\mathbb{P}(M_n \ge \varepsilon) \to 0$ for every $\varepsilon > 0$. Explain why this does not contradict $\mathbb{E}M_n = 1$ or the L^2 convergence theorem of Section 3.

Exercise 2.11 (Efron–Stein)**.** type B Let $Y_1, \dots, Y_N$ be independent, $f(Y_1, \dots, Y_N) \in L^2$, and let Y_k' be an independent copy of Y_k, independent of everything else; write $f^{(k)}$ for f with Y_k replaced by Y_k'. Let ξ_k be the differences of the Doob martingale. (a) Use the freezing lemma to show $\mathbb{E}_{k-1} f = \mathbb{E}_k f^{(k)}$, hence $\xi_k = \mathbb{E}_k[f - f^{(k)}]$. (b) Use rule (viii) to show $\mathbb{E}\xi_k^2 \le \mathbb{E}(f - f^{(k)})^2$. (c) Conclude $\operatorname{Var} f \le \frac{1}{2}\sum_{k \le N} \mathbb{E}(f - f^{(k)})^2$, with equality when $f = \sum_k g_k(Y_k)$.

Exercise 2.12 (SGD on a quadratic)**.** type C Let $\varepsilon_1, \varepsilon_2, \dots$ be i.i.d., centered, with variance σ^2, and consider SGD for $L(\theta) = \frac{1}{2}\mathbb{E}(\theta - \varepsilon)^2$ with step sizes $\eta_k \in (0, 1]$:

$$\theta_{k+1} = \theta_k - \eta_k(\theta_k - \varepsilon_{k+1}) = (1 - \eta_k)\theta_k + \eta_k \varepsilon_{k+1}.$$

(a) Write $\theta_{n+1} = \theta_0 - \sum_{k \le n} \eta_k L'(\theta_k) - M_{n+1}$ as in Example 2.4(d), identifying M_{n+1} explicitly. (b) Derive $\mathbb{E}\theta_{n+1}^2 = (1-\eta_n)^2 \mathbb{E}\theta_n^2 + \eta_n^2 \sigma^2$ and solve it. (c) Show $\mathbb{E}\theta_n^2 \to 0$ if $\eta_n \to 0$ and $\sum \eta_n = \infty$; show that for constant $\eta_n \equiv \eta$, $\mathbb{E}\theta_n^2 \to \eta\sigma^2/(2-\eta)$. (d) Write $\theta_n = \theta_0 \prod_{k<n}(1-\eta_k) + \sum_{k<n} w_{n,k}\varepsilon_{k+1}$ with explicit weights $w_{n,k}$. Which condition on (η_k) makes $\sum_k \eta_k \varepsilon_{k+1}$ converge in L^2? (Theorem 3.10 below.)

3 Decomposition, stopping, convergence

Throughout, M is a martingale with differences ξ_k with respect to the filtration of Section 2.1.

3.1 The Doob decomposition and quadratic variation

Proposition 3.1 (Doob decomposition)**.** *Let $(X_n)_{n \ge 0}$ be adapted. There is exactly one way to write $X_n = X_0 + M_n + A_n$ with M a martingale, A predictable, and $M_0 = A_0 = 0$, namely*

$$A_n = \sum_{k=1}^{n} \big(\mathbb{E}_{k-1} X_k - X_{k-1}\big), \qquad M_n = \sum_{k=1}^{n} \big(X_k - \mathbb{E}_{k-1} X_k\big).$$

X is a submartingale if and only if A is nondecreasing.

Proof. The displayed A is predictable, the displayed M has $\mathbb{E}_{k-1}[X_k - \mathbb{E}_{k-1}X_k] = 0$, and they sum to $X_n - X_0$. If $X_0 + M' + A'$ is another such decomposition, $M - M' = A' - A$ is a predictable martingale, so $M_n - M_n' = \mathbb{E}_{n-1}[M_n - M_n'] = M_{n-1} - M_{n-1}' = \cdots = 0$. The last claim reads off $A_n - A_{n-1} = \mathbb{E}_{n-1}X_n - X_{n-1}$. □

Definition 3.2 (conditional variances, quadratic variation)**.** $\sigma_k^2 := \mathbb{E}_{k-1}[\xi_k^2]$ and $\langle M \rangle_n := \sum_{k=1}^n \sigma_k^2$.

Lemma 3.3. *M_n^2 is a submartingale, $M_n^2 - \langle M \rangle_n$ is a martingale, and $\mathbb{E}\langle M \rangle_n = \mathbb{E}M_n^2 - \mathbb{E}M_0^2$. Equivalently, $\langle M \rangle$ is the predictable part of the Doob decomposition of M^2.*

Proof. $\mathbb{E}_{n-1}[M_n^2] - M_{n-1}^2 = \mathbb{E}_{n-1}[(M_{n-1} + \xi_n)^2] - M_{n-1}^2 = 2M_{n-1}\mathbb{E}_{n-1}\xi_n + \mathbb{E}_{n-1}\xi_n^2 = \sigma_n^2 \ge 0$. □

3.2 Stopping times and optional stopping

Definition 3.4 (stopping time)**.** A random variable τ with values in $\{0,1,2,\dots\}\cup\{\infty\}$ is a *stopping time* if $\mathbf{1}\{\tau \le n\} \in L^2(\mathcal{F}_n)$ for every n, i.e. whether $\tau \le n$ is a function of $Y_1,\dots,Y_n$.

The first entrance time $\min\{n : X_n \in B\}$ of an adapted sequence into a set B is a stopping time; so is $\tau \wedge N = \min(\tau, N)$ for a constant N; the last visit to B is not. For a stopping time τ the *stopped sequence* is $(M_{n\wedge\tau})_{n\ge 0}$.

Proposition 3.5 (stopped martingales)**.** *If M is a martingale (sub-, super-) and τ is a stopping time, then $(M_{n\wedge\tau})_{n\ge0}$ is a martingale (sub-, super-) with differences $\mathbf{1}\{\tau > n\}\xi_{n+1}$.*

Proof. $M_{n\wedge\tau}$ is a function of $Y_1,\dots,Y_n$ and is in L^2 (it is one of $M_0,\dots,M_n$). Since $M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = \mathbf{1}\{\tau > n\}\xi_{n+1}$ and $\mathbf{1}\{\tau > n\} = 1 - \mathbf{1}\{\tau \le n\} \in L^2(\mathcal{F}_n)$ is bounded, rule (iii) gives $\mathbb{E}_n[\mathbf{1}\{\tau > n\}\xi_{n+1}] = \mathbf{1}\{\tau > n\}\mathbb{E}_n\xi_{n+1}$, which is 0 (resp. ≥ 0, ≤ 0). □

Corollary 3.6 (optional stopping, bounded case)**.** *Let $\tau \le N$ be a stopping time bounded by a constant. If M is a martingale then $\mathbb{E}M_\tau = \mathbb{E}M_0$. If X is a submartingale then $\mathbb{E}X_0 \le \mathbb{E}X_\tau \le \mathbb{E}X_N$; for a supermartingale the inequalities reverse.*

Proof. $M_\tau = M_{N\wedge\tau}$ and the stopped martingale has constant mean, so $\mathbb{E}M_\tau = \mathbb{E}M_{0\wedge\tau} = \mathbb{E}M_0$; the same argument gives $\mathbb{E}X_\tau \ge \mathbb{E}X_0$ for a submartingale. For the upper bound,

$$X_N - X_\tau = \sum_{k=1}^{N} \mathbf{1}\{\tau \le k-1\}\,(X_k - X_{k-1}),$$

and each term has $\mathbb{E}\big[\mathbf{1}\{\tau \le k-1\}(X_k - X_{k-1})\big] = \mathbb{E}\big[\mathbf{1}\{\tau \le k-1\}\,\mathbb{E}_{k-1}(X_k - X_{k-1})\big] \ge 0$. □

Boundedness of τ cannot be dropped: for the simple random walk and $\tau = \min\{n : S_n = 1\}$, $\mathbb{E}S_\tau = 1 \ne 0 = \mathbb{E}S_0$. For unbounded τ one applies the corollary to $\tau\wedge n$ and passes to the limit, as follows.

Example 3.7 (gambler's ruin)**.** WORKED IN CLASS Let S be the simple symmetric random walk, a, b positive integers, $\tau = \min\{n : S_n \in \{-b, a\}\}$, and $c = \max(a,b)$.

τ is finite. $S_n^2 - n$ is a martingale (Example 2.4(a)) and $\tau\wedge n$ is bounded, so $\mathbb{E}(\tau\wedge n) = \mathbb{E}S^2_{\tau\wedge n} \le c^2$. Hence $\mathbb{P}(\tau > n) = \mathbb{P}(\tau\wedge n = n) \le \mathbb{E}(\tau\wedge n)/n \le c^2/n \to 0$.

Exit probabilities. $0 = \mathbb{E}S_{\tau\wedge n}$, and $|S_{\tau\wedge n} - S_\tau\mathbf{1}\{\tau\le n\}| \le c\,\mathbf{1}\{\tau > n\}$, so $|\mathbb{E}[S_\tau\mathbf{1}\{\tau\le n\}]| \le c\,\mathbb{P}(\tau > n) \to 0$. With $p = \mathbb{P}(S_\tau = a)$ this gives $pa - (1-p)b = 0$, i.e.

$$\mathbb{P}(S_\tau = a) = \frac{b}{a+b}, \qquad \mathbb{P}(S_\tau = -b) = \frac{a}{a+b}.$$

Expected exit time. $\mathbb{E}(\tau\wedge n) = \mathbb{E}S^2_{\tau\wedge n} \to \mathbb{E}S^2_\tau = a^2\frac{b}{a+b} + b^2\frac{a}{a+b} = ab$, the convergence on the left being $\mathbb{E}(\tau\wedge n) = \sum_{m<n}\mathbb{P}(\tau > m) \uparrow \mathbb{E}\tau$ and on the right as above. So $\mathbb{E}\tau = ab$.

Example 3.8 (Wald's identity)**.** WORKED IN CLASS Let $Y_1, Y_2, \dots$ be i.i.d. with mean μ and finite variance, $S_n = \sum_{k\le n} Y_k$, and τ a stopping time with $\mathbb{E}\tau < \infty$. Then $\mathbb{E}S_\tau = \mu\,\mathbb{E}\tau$. Proof: $S_n - n\mu$ is a martingale, so $\mathbb{E}S_{\tau\wedge n} = \mu\,\mathbb{E}(\tau\wedge n)$ by Corollary 3.6, and the right side tends to $\mu\mathbb{E}\tau$. For the left side write

$$S_\tau - S_{\tau\wedge n} = \sum_{k>n} \mathbf{1}\{\tau \ge k\}(Y_k - \mu) + \mu\,(\tau - n)_+.$$

The first sum has martingale differences (as $\mathbf{1}\{\tau \ge k\} \in L^2(\mathcal{F}_{k-1})$), so by Lemma 2.3 its second moment is $\mathrm{Var}(Y_1)\sum_{k>n}\mathbb{P}(\tau \ge k) = \mathrm{Var}(Y_1)\,\mathbb{E}(\tau-n)_+$; hence $\mathbb{E}|S_\tau - S_{\tau\wedge n}| \le \sqrt{\mathrm{Var}(Y_1)\mathbb{E}(\tau-n)_+} + |\mu|\,\mathbb{E}(\tau-n)_+ \to 0$, and $\mathbb{E}S_{\tau\wedge n} \to \mathbb{E}S_\tau$.

3.3 The maximal inequality and L^2 convergence

Proposition 3.9 (Doob's maximal inequality)**.** *For every n and $u > 0$,*

$$\mathbb{P}\Big(\max_{k\le n}|M_k| \ge u\Big) \le \frac{\mathbb{E}M_n^2}{u^2}, \qquad \text{and} \qquad \mathbb{E}\max_{k\le n} M_k^2 \le 4\,\mathbb{E}M_n^2.$$

Proof. Let $\tau = \min\{k \le n : |M_k| \ge u\}$, with $\tau = n$ if no such k exists; $\tau \le n$ is a stopping time. M^2 is a submartingale (Lemma 3.3), so by Corollary 3.6

$$\mathbb{E}M_n^2 \ge \mathbb{E}M_\tau^2 \ge \mathbb{E}\big[M_\tau^2 \mathbf{1}\{\max_{k\le n}|M_k| \ge u\}\big] \ge u^2\,\mathbb{P}\Big(\max_{k\le n}|M_k| \ge u\Big).$$

The second inequality is Exercise 3.13. □

Theorem 3.10 (L^2 convergence)**.** *Suppose $\sum_{k\ge1}\mathbb{E}\xi_k^2 < \infty$, equivalently $\sup_n \mathbb{E}M_n^2 < \infty$. Then:*

(i) there is $M_\infty \in L^2$ with $\mathbb{E}(M_n - M_\infty)^2 = \sum_{k>n}\mathbb{E}\xi_k^2 \to 0$, and $\mathbb{E}_n M_\infty = M_n$ for every n;

(ii) for every n and $u > 0$, $\mathbb{P}\Big(\sup_{k\ge n}|M_k - M_n| \ge u\Big) \le \dfrac{1}{u^2}\displaystyle\sum_{k>n}\mathbb{E}\xi_k^2$.

Proof. (i) $\mathbb{E}(M_n - M_m)^2 = \sum_{k=n+1}^m \mathbb{E}\xi_k^2$ for $n < m$ (Lemma 2.3), so (M_n) is Cauchy in L^2 and converges to some M_∞ by completeness, with $\mathbb{E}(M_n - M_\infty)^2 = \lim_m \mathbb{E}(M_n - M_m)^2$. Since $\mathbb{E}_n M_m = M_n$ for $m \ge n$ and $\mathbb{E}_n$ is a contraction (rule (viii)), $\mathbb{E}_n M_\infty = \lim_m \mathbb{E}_n M_m = M_n$. (ii) For fixed n, $(M_k - M_n)_{k\ge n}$ is a martingale with differences ξ_k, $k > n$. Proposition 3.9 with horizon m gives $\mathbb{P}(\max_{n\le k\le m}|M_k - M_n| \ge u) \le u^{-2}\sum_{k=n+1}^m \mathbb{E}\xi_k^2$; let $m \to \infty$. □

Statement (ii) controls the entire future of the path from time n on; it is the form in which convergence will be used.

Example 3.11 (weighted sums)**.** Let $\varepsilon_1, \varepsilon_2, \dots$ be martingale differences with $\mathbb{E}_{k-1}\varepsilon_k^2 \le 1$ and let a_k be predictable and bounded with $\sum_k \mathbb{E}a_k^2 < \infty$. Then $M_n = \sum_{k\le n} a_k\varepsilon_k$ is a martingale with $\sigma_k^2 = a_k^2\mathbb{E}_{k-1}\varepsilon_k^2 \le a_k^2$, and Theorem 3.10 applies. In particular, for deterministic η_k,

$$\sum_k \eta_k\varepsilon_k \ \text{ converges in } L^2 \ \text{ whenever } \ \sum_k \eta_k^2 < \infty.$$

This is the Robbins–Monro condition on the step sizes of SGD (Lecture 16).

Example 3.12 (random series)**.** WORKED IN CLASS Let ε_k be i.i.d. signs. $\sum_k \varepsilon_k/k$ converges in L^2, since $\sum k^{-2} < \infty$. $\sum_k \varepsilon_k/\sqrt{k}$ does not: $\mathbb{E}M_n^2 = \sum_{k\le n} 1/k \to \infty$, and $\mathbb{E}(M_m - M_n)^2 = \sum_{n<k\le m} 1/k$ does not tend to zero as $n, m \to \infty$. For Pólya's urn (Example 2.6), $\sum_n \mathbb{E}\xi_n^2 < \infty$, so $F_n \to F_\infty$ in L^2 with $\mathbb{E}F_\infty = r/(r+b)$.

Exercises for Lecture 13

Exercise 3.13. TYPE B Let $M_n^* = \max_{k\le n}|M_k|$. (a) Show $|M|$ is a submartingale (rule (vi)). (b) Refine the proof of Proposition 3.9 to $u\,\mathbb{P}(M_n^* \ge u) \le \mathbb{E}\big[|M_n|\mathbf{1}\{M_n^* \ge u\}\big]$. (c) Use $\mathbb{E}Z^2 = \int_0^\infty 2u\mathbb{P}(Z\ge u)\,du$ and Cauchy–Schwarz to conclude $\mathbb{E}(M_n^*)^2 \le 4\mathbb{E}M_n^2$.

Exercise 3.14. TYPE A Compute $\langle M\rangle_n$ for: the simple random walk; the exponential martingale of Example 2.4(a); the Doob martingale of $f = \sum_{k\le N} Y_k$ with independent Y_k; the likelihood ratio of Example 2.5.

Exercise 3.15. TYPE B Let S be the simple symmetric random walk and $\tau_1 = \min\{n : S_n = 1\}$. (a) Use Example 3.7 with $a = 1$ and $b \to \infty$ to show $\mathbb{P}(\tau_1 < \infty) = 1$. (b) Show $\mathbb{E}\tau_1 = \infty$ (if it were finite, Wald's identity would give $1 = \mathbb{E}S_{\tau_1} = 0$). (c) Apply Corollary 3.6 to the martingale $e^{\lambda S_n}/(\cosh\lambda)^n$ at $\tau_1 \wedge n$ and let $n\to\infty$ to compute $\mathbb{E}\big[(\cosh\lambda)^{-\tau_1}\big]$ for $\lambda > 0$.

Exercise 3.16 (Pólya's urn, finished)**.** TYPE B Continue Example 2.6. (a) For $r = b = 1$ show by induction that R_n is uniform on $\{1,\dots,n+1\}$, so the L^2 limit F_∞ is uniform on $[0,1]$. (b) Compute $\operatorname{Var} F_n$ for general r, b from the conditional variances and identify $\operatorname{Var} F_\infty$. (c) Show $\mathbb{E}[F_\infty \mid \mathcal{F}_n] = F_n$ and interpret: the urn's eventual composition is random, but its best guess at time n is the current fraction.

Exercise 3.17 (branching processes)**.** TYPE C Let $Z_0 = 1$ and $Z_{n+1} = \sum_{i=1}^{Z_n} \zeta_{n+1,i}$, where the $\zeta_{n,i}$ are i.i.d. nonnegative integers with mean m and variance s^2, independent of the past. (a) Use the freezing lemma to show $\mathbb{E}_n Z_{n+1} = mZ_n$, so $W_n = Z_n/m^n$ is a martingale. (b) Use Definition 1.4 to show $\operatorname{Var} Z_{n+1} = s^2m^n + m^2\operatorname{Var} Z_n$, hence $\operatorname{Var} W_{n+1} = \operatorname{Var} W_n + s^2/m^{n+2}$. (c) For $m > 1$ conclude $\sup_n \mathbb{E}W_n^2 < \infty$, so $W_n \to W_\infty$ in L^2 with $\mathbb{E}W_\infty = 1$ and $\operatorname{Var} W_\infty = s^2/(m(m-1))$. (d) Simulate for Poisson offspring with $m = 1.5$ and plot W_n along several runs.

4 The martingale central limit theorem

Lectures 3–4 proved the CLT for independent summands by swapping them one at a time for Gaussians. For martingale differences the conditional variance of each step is random, so the Gaussian part of the hybrid is taken to carry the *remaining* conditional variance.

4.1 Statement

Fix k and martingale differences $\xi_1,\dots,\xi_k$ with $\mathbb{E}|\xi_i|^3 < \infty$, conditional variances $\sigma_i^2 = \mathbb{E}_{i-1}\xi_i^2$, and

$$S_j = \sum_{i\le j}\xi_i, \qquad V_j = \sum_{i\le j}\sigma_i^2 = \langle S\rangle_j, \qquad S_0 = V_0 = 0.$$

Theorem 4.1 (martingale CLT, quantitative form)**.** *Suppose $V_k \le 1$. Then for every $f \in C^4(\mathbb{R})$ with f', f'', f''', f'''' bounded, and $g \sim \mathcal{N}(0,1)$,*

$$\big|\mathbb{E}f(S_k) - \mathbb{E}f(g)\big| \;\le\; \frac{\|f'''\|_\infty}{6}\sum_{i=1}^k \mathbb{E}|\xi_i|^3 \;+\; \frac{\|f''''\|_\infty}{8}\sum_{i=1}^k \mathbb{E}\,\sigma_i^4 \;+\; \frac{\|f''\|_\infty}{2}\,\mathbb{E}\big(1 - V_k\big).$$

Since $\sum_i \sigma_i^4 \le V_k \max_i \sigma_i^2 \le \max_i \sigma_i^2$, the second term is at most $\frac{1}{8}\|f''''\|_\infty \mathbb{E}\max_i \sigma_i^2$.

Corollary 4.2 (martingale CLT)**.** *For each n let $\xi_{n,1},\dots,\xi_{n,k_n}$ be martingale differences (for a filtration that may depend on n) with conditional variances $\sigma_{n,i}^2$ and $V_{n,k_n}=\sum_i \sigma_{n,i}^2 \le 1$. If*

$$\sum_i \mathbb{E}|\xi_{n,i}|^3 \to 0, \qquad \mathbb{E}\max_i \sigma_{n,i}^2 \to 0, \qquad \mathbb{E}\big(1-V_{n,k_n}\big)\to 0,$$

then $\sum_i \xi_{n,i} \Rightarrow \mathcal{N}(0,1)$.

Proof. By Theorem 4.1, $\mathbb{E}f(S_{n,k_n}) \to \mathbb{E}f(g)$ for every smooth f with bounded derivatives of all orders, and such test functions determine convergence in distribution (Lectures 3–4). □

4.2 Proof

Lemma 4.3 (the heat semigroup)**.** *For f as in Theorem 4.1 and $g\sim\mathcal{N}(0,1)$ put $\phi(x,v)=\mathbb{E}f(x+\sqrt{v}\,g)$, $x\in\mathbb{R}$, $v\ge 0$. Then $\phi(x,0)=f(x)$, $\phi(0,1)=\mathbb{E}f(g)$, $\partial_x^j\phi(x,v)=\mathbb{E}f^{(j)}(x+\sqrt{v}g)$ so that $\|\partial_x^j\phi\|_\infty\le\|f^{(j)}\|_\infty$ ($j\le 4$), and for $v>0$*

$$\partial_v\phi=\tfrac12\,\partial_{xx}\phi.$$

Consequently $\phi(x,v)-\phi(x,v')=\frac12\int_{v'}^{v}\partial_{xx}\phi(x,s)\,ds$ for $0\le v'<v$.

Proof. Differentiation under $\mathbb{E}$ is justified by the bounded derivatives. For the heat equation apply Stein's identity $\mathbb{E}[g\,h(g)]=\mathbb{E}h'(g)$ (Lectures 3–4) to $h(y)=f'(x+\sqrt{v}\,y)$:

$$\partial_v\phi(x,v)=\mathbb{E}\Big[f'(x+\sqrt{v}g)\frac{g}{2\sqrt{v}}\Big]=\frac{1}{2\sqrt{v}}\mathbb{E}h'(g)=\frac12\mathbb{E}f''(x+\sqrt{v}g).$$

The integral formula holds for $v'>0$ by the fundamental theorem of calculus; let $v'\downarrow 0$ using $|\phi(x,v')-\phi(x,0)|\le\|f'\|_\infty\sqrt{v'}\,\mathbb{E}|g|$. □

Proof of Theorem 4.1. Since $\phi(S_0,1-V_0)=\mathbb{E}f(g)$ and $\phi(S_k,0)=f(S_k)$,

$$\mathbb{E}f(S_k)-\mathbb{E}f(g)=\mathbb{E}\big[\phi(S_k,0)-\phi(S_k,1-V_k)\big]+\sum_{i=1}^{k}\mathbb{E}\big[\phi(S_i,1-V_i)-\phi(S_{i-1},1-V_{i-1})\big].$$

First term. By the integral formula, $|\phi(x,0)-\phi(x,v)|\le\frac{v}{2}\|f''\|_\infty$; so the first term is at most $\frac{\|f''\|_\infty}{2}\mathbb{E}(1-V_k)$ in absolute value.

Step i. Write $x=S_{i-1}$, $v=1-V_{i-1}$, $\sigma^2=\sigma_i^2$, $\xi=\xi_i$; then $x,v,\sigma^2\in L^2(\mathcal{F}_{i-1})$ and $v-\sigma^2=1-V_i\ge 0$. Split

$$\phi(x+\xi,v-\sigma^2)-\phi(x,v)=\underbrace{\phi(x+\xi,v-\sigma^2)-\phi(x,v-\sigma^2)}_{I}+\underbrace{\phi(x,v-\sigma^2)-\phi(x,v)}_{II}.$$

Taylor in x: $I=\xi\,\partial_x\phi(x,v-\sigma^2)+\frac{\xi^2}{2}\partial_{xx}\phi(x,v-\sigma^2)+R$ with $|R|\le\frac{\|f'''\|_\infty}{6}|\xi|^3$. The coefficients are bounded and in $L^2(\mathcal{F}_{i-1})$, so by rule (iii)

$$\mathbb{E}_{i-1}I=\frac{\sigma^2}{2}\,\partial_{xx}\phi(x,v-\sigma^2)+\mathbb{E}_{i-1}R.$$

By the integral formula,

$$II = -\frac{1}{2}\int_{v-\sigma^2}^{v} \partial_{xx}\phi(x,s)\,ds = -\frac{\sigma^2}{2}\,\partial_{xx}\phi(x,v-\sigma^2) - \frac{1}{2}\int_{v-\sigma^2}^{v} \big[\partial_{xx}\phi(x,s) - \partial_{xx}\phi(x,v-\sigma^2)\big]\,ds,$$

and the leading terms of $\mathbb{E}_{i-1}I$ and II cancel. Since $\partial_s\partial_{xx}\phi = \frac{1}{2}\partial_x^4\phi$ is bounded by $\frac{1}{2}\|f''''\|_\infty$, the integrand is at most $\frac{1}{2}\|f''''\|_\infty(s-v+\sigma^2)$ in absolute value, and the integral term is at most $\frac{1}{4}\|f''''\|_\infty\int_0^{\sigma^2} t\,dt = \frac{\|f''''\|_\infty}{8}\sigma^4$. Hence

$$\big|\mathbb{E}_{i-1}[I+II]\big| \le \frac{\|f'''\|_\infty}{6}\,\mathbb{E}_{i-1}|\xi_i|^3 + \frac{\|f''''\|_\infty}{8}\,\sigma_i^4.$$

Take expectations and sum over i. □

Remark 4.4 (removing $V_k \le 1$). $\{V_i \le 1\}$ is decided at time $i-1$ (as $V_i = V_{i-1} + \sigma_i^2$ with $\sigma_i^2 \in L^2(\mathcal{F}_{i-1})$), so $\xi_i' = \xi_i \mathbf{1}\{V_i \le 1\}$ are martingale differences with conditional variances $\sigma_i^2\mathbf{1}\{V_i \le 1\}$ and accumulated variance $V_k' \le 1$. By Lemma 2.3,

$$\mathbb{E}\big(S_k - S_k'\big)^2 = \mathbb{E}\sum_{i:\,V_i>1}\sigma_i^2 \le \mathbb{E}\big(V_k - 1\big)_+ + \mathbb{E}\max_i \sigma_i^2.$$

So Theorem 4.1 applied to S_k' controls S_k up to this error, and Corollary 4.2 holds with $V_{n,k_n} \le 1$ replaced by $\mathbb{E}|V_{n,k_n} - 1| \to 0$.

Remark 4.5 (independent summands). If the ξ_i are independent with variances σ_i^2 summing to one, then $V_k = 1$ and the bound reads $|\mathbb{E}f(S_k) - \mathbb{E}f(g)| \le \frac{\|f'''\|_\infty}{6}\sum\mathbb{E}|\xi_i|^3 + \frac{\|f''''\|_\infty}{8}\sum\sigma_i^4$. The hybrid $\sum_{j\le i}\xi_j + \sum_{j>i} g_j$ of Lectures 3–4 has become $S_i + \sqrt{1-V_i}\,g$.

4.3 Examples

Example 4.6 (i.i.d. summands). WORKED IN CLASS Let $X_1,\dots,X_n$ be i.i.d., centered, with unit variance and $\mathbb{E}|X_1|^3 < \infty$, and $\xi_i = X_i/\sqrt{n}$. Then $\sigma_i^2 = 1/n$, $V_n = 1$, $\sum\mathbb{E}|\xi_i|^3 = \mathbb{E}|X_1|^3/\sqrt{n}$, $\sum\sigma_i^4 = 1/n$, and Theorem 4.1 gives $|\mathbb{E}f(S_n) - \mathbb{E}f(g)| \le \frac{\|f'''\|_\infty\mathbb{E}|X_1|^3}{6\sqrt{n}} + \frac{\|f''''\|_\infty}{8n}$.

Example 4.7 (adaptively collected data). WORKED IN CLASS Let $\varepsilon_1,\varepsilon_2,\dots$ be i.i.d., centered, with unit variance and $\mathbb{E}|\varepsilon_1|^3 < \infty$, adapted to a filtration $(\mathcal{F}_i)$ with ε_i independent of $\mathcal{F}_{i-1}$, and let a_i be predictable with $|a_i| \le A$. Then $\xi_i = a_i\varepsilon_i$ are martingale differences with

$$\sigma_i^2 = a_i^2, \qquad \mathbb{E}_{i-1}|\xi_i|^3 = |a_i|^3\,\mathbb{E}|\varepsilon_1|^3, \qquad V_k = \sum_{i\le k} a_i^2.$$

Let $v_k \to \infty$ be deterministic with $\mathbb{E}|V_k/v_k - 1| \to 0$. The normalized differences $a_i\varepsilon_i/\sqrt{v_k}$ satisfy $\sum_i \mathbb{E}|\cdot|^3 \le A\,\mathbb{E}|\varepsilon_1|^3\,\mathbb{E}V_k/v_k^{3/2} \to 0$ and $\max_i\sigma_i^2/v_k \le A^2/v_k \to 0$, so by Corollary 4.2 and Remark 4.4

$$\frac{1}{\sqrt{v_k}}\sum_{i\le k} a_i\varepsilon_i \;\Rightarrow\; \mathcal{N}(0,1).$$

This covers least squares with covariates a_i chosen as a function of past responses (bandits, active learning, sequential experiments): Exercises 4.10 and 4.11.

Exercises for Lecture 14

Exercise 4.8. type A Let ξ_i be independent, centered, with variances σ_i^2, $\sum_{i\le k}\sigma_i^2 = 1$. Deduce Lindeberg's theorem of Lectures 3–4 (under a third-moment condition) from Theorem 4.1, and compare the error bounds.

Exercise 4.9. type A Verify directly that $\phi(x,v) = \mathbb{E}f(x+\sqrt{v}g)$ solves $\partial_v\phi = \tfrac12\partial_{xx}\phi$ when $f(x) = e^{i\theta x}$, and when $f(x) = x^4$.

Exercise 4.10 (a two-armed bandit)**.** type B Two arms give rewards $\mu_1+\varepsilon$ and $\mu_2+\varepsilon$, with ε as in Example 4.7. At each round the player pulls the arm with the larger current empirical mean with probability $1-\delta$, and the other arm with probability $\delta \in (0,\tfrac12)$, using an independent coin. Let $N_1(n)$ be the number of pulls of arm 1 in the first n rounds and $\hat\mu_1(n)$ its empirical mean. (a) Show $N_1(n)$ dominates a $\mathrm{Bin}(n,\delta)$ variable, so $N_1(n)\to\infty$. (b) Write $N_1(n)(\hat\mu_1(n)-\mu_1) = \sum_{i\le n} a_i\varepsilon_i$ with $a_i \in \{0,1\}$ predictable. (c) Assuming $N_1(n)/n \to p_1 \in (0,1)$ in L^2, show $\sqrt{N_1(n)}\,(\hat\mu_1(n)-\mu_1) \Rightarrow \mathcal{N}(0,1)$.

Exercise 4.11 (least squares for an autoregression)**.** type C Let $X_0 = 0$, $X_{k+1} = \rho X_k + \varepsilon_{k+1}$ with $|\rho|<1$ and ε_k i.i.d. $\mathcal{N}(0,\sigma^2)$. The least squares estimate of ρ from $X_0,\dots,X_n$ satisfies

$$\hat\rho_n - \rho = \frac{\sum_{k<n} X_k\varepsilon_{k+1}}{\sum_{k<n} X_k^2}.$$

(a) Show the numerator M_n is a martingale with $\sigma_{k+1}^2 = \sigma^2 X_k^2$. (b) Show $\mathbb{E}X_k^2 \to \sigma^2/(1-\rho^2)$ and $\mathrm{Var}\left(\frac1n\sum_{k<n}X_k^2\right) = O(1/n)$ (Wick's theorem for the fourth moments; $\mathrm{Cov}(X_j^2,X_k^2)$ decays geometrically in $|j-k|$), so $\frac1n\sum_{k<n}X_k^2 \to \sigma^2/(1-\rho^2)$ in L^2. (c) Apply Corollary 4.2 with Remark 4.4 to $M_n/\sqrt{n\sigma^4/(1-\rho^2)}$ and conclude $\sqrt{n}(\hat\rho_n-\rho) \Rightarrow \mathcal{N}(0,1-\rho^2)$ (quote that dividing by a sequence converging in probability to a positive constant does not change a distributional limit). (d) Simulate for $\rho = 0.9$, $n = 2000$, 10^3 runs, and compare with the predicted Gaussian.

Exercise 4.12 (SGD on a quadratic: fluctuations)**.** type C In Exercise 2.12 take $\eta_k = \eta/k$ with $\eta > \tfrac12$ and $\mathbb{E}|\varepsilon_1|^3 < \infty$. (a) Show the weights satisfy $w_{n,k} = \frac{\eta}{k}\left(\frac{k}{n}\right)^{\eta}(1+o(1))$ uniformly in $k \le n$. (b) Show $n\,\mathrm{Var}\,\theta_n \to \eta^2\sigma^2/(2\eta-1)$. (c) Apply Theorem 4.1 (the summands are independent, so V_k is deterministic) to prove $\sqrt{n}\,\theta_n \Rightarrow \mathcal{N}\big(0,\eta^2\sigma^2/(2\eta-1)\big)$. What changes for $\eta < \tfrac12$?

5 Concentration

Throughout, M is a martingale with differences ξ_k and conditional variances σ_k^2.

5.1 Azuma–Hoeffding

Lemma 5.1 (Hoeffding's lemma)**.** *Let $\mathbb{E}\xi = 0$ and $a \le \xi \le a+c$ for constants a, c. Then $\mathbb{E}e^{\lambda\xi} \le e^{\lambda^2c^2/8}$ for all $\lambda \in \mathbb{R}$. Conditionally: if $\mathbb{E}_{k-1}\xi = 0$ and $A \le \xi \le A+c$ with $A \in L^2(\mathcal{F}_{k-1})$ and c a constant, then $\mathbb{E}_{k-1}e^{\lambda\xi} \le e^{\lambda^2c^2/8}$.*

Proof. We prove the case $|\xi| \le c$ with the constant $e^{\lambda^2c^2/2}$; the sharp constant is Exercise 5.13. By convexity of $x \mapsto e^{\lambda x}$ on $[-c,c]$,

$$e^{\lambda\xi} \le \frac{c-\xi}{2c}e^{-\lambda c} + \frac{c+\xi}{2c}e^{\lambda c},$$

so $\mathbb{E}e^{\lambda\xi} \le \cosh(\lambda c) \le e^{\lambda^2 c^2/2}$, the last step by comparing Taylor series ($(2k)! \ge 2^k k!$). The conditional version is the same computation with $\mathbb{E}_{k-1}$ (rules (iii), (vi)). □

Theorem 5.2 (Azuma–Hoeffding). *Suppose $A_k \le \xi_k \le A_k + c_k$ with A_k predictable and c_k constants. Then for every $u > 0$,*

$$\mathbb{P}\big(M_n - M_0 \ge u\big) \le \exp\Big(-\frac{2u^2}{\sum_{k\le n} c_k^2}\Big),$$

and the same bound holds for $\mathbb{P}(M_n - M_0 \le -u)$.

Proof. By rules (ii)–(iv) and Lemma 5.1,

$$\mathbb{E}e^{\lambda(M_n - M_0)} = \mathbb{E}\Big[e^{\lambda(M_{n-1}-M_0)}\,\mathbb{E}_{n-1}e^{\lambda\xi_n}\Big] \le e^{\lambda^2 c_n^2/8}\,\mathbb{E}e^{\lambda(M_{n-1}-M_0)} \le \cdots \le \exp\Big(\frac{\lambda^2}{8}\sum_{k\le n} c_k^2\Big).$$

Chernoff's bound $\mathbb{P}(M_n - M_0 \ge u) \le e^{-\lambda u}\mathbb{E}e^{\lambda(M_n - M_0)}$ with $\lambda = 4u/\sum c_k^2$ gives the claim; apply it to $-M$ for the lower tail. □

For the simple random walk, $\mathbb{P}(S_n \ge u) \le e^{-u^2/2n}$.

5.2 McDiarmid's inequality

Theorem 5.3 (bounded differences). *Let $Y_1, \dots, Y_N$ be independent and let f be bounded with*

$$\big|f(y_1,\dots,y_N) - f(y_1,\dots,y_{k-1},y_k',y_{k+1},\dots,y_N)\big| \le c_k \qquad \text{for all arguments and all } k.$$

Then for every $u > 0$, $\mathbb{P}\big(f(Y) - \mathbb{E}f(Y) \ge u\big) \le \exp\big(-2u^2/\sum_{k\le N} c_k^2\big)$, and the same for the lower tail.

Proof. Let $M_k = \mathbb{E}_k f$ be the Doob martingale. By the freezing lemma, $M_k = g_k(Y_1,\dots,Y_k)$ with $g_k(y_1,\dots,y_k) = \mathbb{E}f(y_1,\dots,y_k,Y_{k+1},\dots,Y_N)$, and $|g_k(\dots,y,\dots) - g_k(\dots,y',\dots)| \le c_k$ when the arguments differ only in the k-th place. Hence, given $Y_1,\dots,Y_{k-1}$, the difference $\xi_k = g_k(Y_1,\dots,Y_k) - g_{k-1}(Y_1,\dots,Y_{k-1})$ lies in $[A_k, A_k + c_k]$ with $A_k = \inf_y g_k(Y_1,\dots,Y_{k-1},y) - g_{k-1}(Y_1,\dots,Y_{k-1})$ predictable. Apply Theorem 5.2 to $M_N - M_0 = f - \mathbb{E}f$. □

Example 5.4 (Hoeffding's inequality). For $f = \frac{1}{n}\sum_{i\le n} Y_i$ with $Y_i \in [0,1]$, $c_i = 1/n$ and $\mathbb{P}(\bar{Y}_n - \mathbb{E}\bar{Y}_n \ge u) \le e^{-2nu^2}$.

Example 5.5 (the generalization gap). WORKED IN CLASS Let $z_1,\dots,z_n$ be i.i.d., $\mathcal{H}$ a class of predictors, $\ell(h,z) \in [0,1]$ a loss, $L(h) = \mathbb{E}\ell(h,z)$, $\hat{L}_n(h) = \frac{1}{n}\sum_i \ell(h,z_i)$, and

$$Z = \sup_{h\in\mathcal{H}} \big(L(h) - \hat{L}_n(h)\big).$$

Changing one z_i changes each $\hat{L}_n(h)$, hence Z, by at most $1/n$. Theorem 5.3 gives, for any class $\mathcal{H}$,

$$\mathbb{P}\big(Z \ge \mathbb{E}Z + u\big) \le e^{-2nu^2}.$$

Proposition 5.6 (symmetrization). *With $\varepsilon_1,\dots,\varepsilon_n$ independent uniform signs independent of the data,*

$$\mathbb{E}Z \le 2\,\mathbb{E}\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_{i\le n} \varepsilon_i\,\ell(h,z_i).$$

Proof. Let $z_1', \dots, z_n'$ be an independent copy of the data and $\mathbb{E}'$ the expectation over the copy, so $L(h) = \mathbb{E}'\hat{L}_n'(h)$. Then

$$\mathbb{E}Z = \mathbb{E}\sup_h \mathbb{E}'\big[\hat{L}_n'(h) - \hat{L}_n(h)\big] \le \mathbb{E}\sup_h \big[\hat{L}_n'(h) - \hat{L}_n(h)\big] = \mathbb{E}\sup_h \frac{1}{n}\sum_i \big(\ell(h, z_i') - \ell(h, z_i)\big).$$

The joint law of the family $\big(\ell(h, z_i') - \ell(h, z_i)\big)_{h,i}$ is unchanged if the i-th term is multiplied by ε_i, since z_i and z_i' are exchangeable. Hence the right side equals $\mathbb{E}\sup_h \frac{1}{n}\sum_i \varepsilon_i(\ell(h, z_i') - \ell(h, z_i)) \le 2\mathbb{E}\sup_h \frac{1}{n}\sum_i \varepsilon_i \ell(h, z_i)$. □

Corollary 5.7 (finite classes)**.** *If $|\mathcal{H}| = m$ then $\mathbb{E}Z \le 2\sqrt{2\log m/n}$, hence $\mathbb{P}\big(Z \ge 2\sqrt{2\log m/n} + u\big) \le e^{-2nu^2}$.*

Proof. Conditionally on the data, $X_h = \frac{1}{n}\sum_i \varepsilon_i \ell(h, z_i)$ satisfies $\mathbb{E}e^{\lambda X_h} = \prod_i \cosh(\lambda \ell(h, z_i)/n) \le e^{\lambda^2/2n}$. The proof of Proposition 2.2 of Lectures 9–11 uses only this bound, so $\mathbb{E}\max_h X_h \le \sqrt{2\log m/n}$. □

For infinite classes the right side of Proposition 5.6 is bounded by chaining, as in Dudley's inequality, with the Rademacher process in place of the Gaussian one: chaining bounds the mean of the supremum, McDiarmid its fluctuations.

5.3 Ville's inequality

Proposition 5.8 (Ville)**.** *Let Z be a nonnegative supermartingale. Then for every $a > 0$,*

$$\mathbb{P}\Big(\sup_{n \ge 0} Z_n \ge a\Big) \le \frac{\mathbb{E}Z_0}{a}.$$

Proof. Let $\tau = \min\{n : Z_n \ge a\}$. By Corollary 3.6 for the supermartingale Z at the bounded time $\tau \wedge n$, and nonnegativity,

$$\mathbb{E}Z_0 \;\ge\; \mathbb{E}Z_{\tau\wedge n} \;\ge\; \mathbb{E}\big[Z_{\tau\wedge n}\mathbf{1}\{\tau \le n\}\big] \;\ge\; a\,\mathbb{P}(\tau \le n).$$

Let $n \to \infty$. □

A nonnegative supermartingale with $Z_0 = 1$ exceeds a at some time with probability at most $1/a$, uniformly over all stopping rules; this is the inequality behind sequential tests and confidence sequences.

5.4 Freedman's inequality

Lemma 5.9. *Let $\psi_c(\lambda) = (e^{\lambda c} - 1 - \lambda c)/c^2$, $\lambda > 0$. If $\mathbb{E}_{k-1}\xi = 0$, $\xi \le c$ and $\sigma^2 = \mathbb{E}_{k-1}\xi^2$, then*

$$\mathbb{E}_{k-1}e^{\lambda\xi} \le 1 + \psi_c(\lambda)\,\sigma^2 \le \exp\big(\psi_c(\lambda)\sigma^2\big).$$

Proof. $h(x) = (e^x - 1 - x)/x^2$ ($h(0) = \frac{1}{2}$) is increasing on $\mathbb{R}$ (Exercise 5.14), so $e^x \le 1 + x + x^2 h(\lambda c)$ for $x \le \lambda c$. With $x = \lambda\xi$ and $\lambda^2 h(\lambda c) = \psi_c(\lambda)$, $e^{\lambda\xi} \le 1 + \lambda\xi + \psi_c(\lambda)\xi^2$; take $\mathbb{E}_{k-1}$ (rule (vi)). □

Corollary 5.10. *Let $M_0 = 0$ and $\xi_k \le c$ for all k. For every $\lambda > 0$, $Z_n = \exp\big(\lambda M_n - \psi_c(\lambda)\langle M\rangle_n\big)$ is a nonnegative supermartingale with $Z_0 = 1$.*

Proof. $Z_n \le e^{\lambda n c}$ is bounded, and $\langle M\rangle_n$ is predictable, so $\mathbb{E}_{n-1} Z_n = Z_{n-1} e^{-\psi_c(\lambda)\sigma_n^2}\mathbb{E}_{n-1} e^{\lambda\xi_n} \le Z_{n-1}$ by Lemma 5.9. □

Theorem 5.11 (Freedman's inequality)**.** *Let $M_0 = 0$ and $\xi_k \le c$ for all k. Then for all $u, v > 0$,*

$$\mathbb{P}\Big(\exists\, n:\ M_n \ge u \ \text{ and } \ \langle M\rangle_n \le v\Big) \le \exp\Big(-\frac{u^2}{2(v + cu/3)}\Big).$$

Proof. On the event, $Z_n \ge \exp(\lambda u - \psi_c(\lambda)v)$ for the n in question, so by Ville's inequality

$$\mathbb{P}(\cdots) \le \mathbb{P}\Big(\sup_n Z_n \ge e^{\lambda u - \psi_c(\lambda)v}\Big) \le \exp\big(-\lambda u + \psi_c(\lambda)v\big).$$

For $0 < \lambda < 3/c$, $\psi_c(\lambda) \le \frac{\lambda^2}{2(1-\lambda c/3)}$ (Exercise 5.14). Take $\lambda = u/(v + cu/3) < 3/c$; then $1 - \lambda c/3 = v/(v + cu/3)$ and the exponent is

$$-\frac{u^2}{v + cu/3} + \frac{u^2}{(v + cu/3)^2}\cdot\frac{v + cu/3}{2} = -\frac{u^2}{2(v + cu/3)}.$$ □

Two properties distinguish Theorem 5.11 from Theorem 5.2: the bound depends on the realized quadratic variation $\langle M\rangle_n$ rather than on $\sum c_k^2$ (for $v \gg cu$ it is the Gaussian tail $e^{-u^2/2v}$; Bernstein's inequality for independent sums is the case of deterministic σ_k^2), and it holds simultaneously for all n.

Example 5.12 (state-dependent noise)**.** WORKED IN CLASS Let $M_n = \sum_{k\le n}\eta_k\xi_k$ with $|\xi_k| \le c$ and $\sigma_k^2 = s^2(\theta_{k-1})$ for some adapted sequence (θ_k) and a function $s^2 \le c^2$. Azuma gives $\mathbb{P}(M_n \ge u) \le \exp(-u^2/(2c^2\sum_{k\le n}\eta_k^2))$. Freedman gives, for every v,

$$\mathbb{P}\Big(\exists n:\ M_n \ge u,\ \sum_{k\le n}\eta_k^2 s^2(\theta_{k-1}) \le v\Big) \le \exp\Big(-\frac{u^2}{2(v + \eta_{\max}cu/3)}\Big),$$

where $\eta_{\max} = \max_k \eta_k$. When the path (θ_k) spends its time where s^2 is small, v can be taken much smaller than $c^2\sum\eta_k^2$. For SGD this is the regime where the gradient noise vanishes near the minimizer (Lecture 17).

QUOTED · Gaussian concentration

Let $g \sim \mathcal{N}(0, I_d)$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be L-Lipschitz for the Euclidean norm. Then $\mathbb{P}\big(f(g) - \mathbb{E}f(g) \ge u\big) \le e^{-u^2/2L^2}$ for all $u > 0$. This is the Gaussian analogue of Theorem 5.3, with L in place of $(\sum c_k^2)^{1/2}$ and no bounded-differences hypothesis; it can be proved by the interpolation of Lecture 11.

Exercises for Lecture 15

Exercise 5.13. TYPE A (a) Prove Lemma 5.1 with the constant $e^{\lambda^2 c^2/8}$: write ξ as a convex combination of the endpoints a, $a + c$ to get $\mathbb{E}e^{\lambda\xi} \le e^{\varphi(\lambda c)}$ with $\varphi(t) = -pt + \log(1 - p + pe^t)$, $p = -a/c$, and show $\varphi(0) = \varphi'(0) = 0$, $\varphi'' \le \frac{1}{4}$. (b) For m fixed predictors and a loss in $[0,1]$, bound $\mathbb{P}(\max_{j\le m}|\hat{L}_n(h_j) - L(h_j)| \ge u)$ by a union bound over Example 5.4, and compare with Corollary 5.7.

Exercise 5.14. type A (a) Show $h(x) = (e^x - 1 - x)/x^2$ is increasing on $\mathbb{R}$. (Put $k(x) = x^3 h'(x) = (x-2)e^x + x + 2$; $k'(x) = (x-1)e^x + 1$ and $k''(x) = xe^x$, so $k' \geq 0$, k is increasing with $k(0) = 0$, and $h'(x) = k(x)/x^3 \geq 0$ on both half-lines.) (b) Show $\psi_c(\lambda) \leq \frac{\lambda^2}{2(1-\lambda c/3)}$ for $0 < \lambda < 3/c$, using $k! \geq 2 \cdot 3^{k-2}$.

Exercise 5.15 (linear predictors)**.** type B Let $x_1, \dots, x_n \in \mathbb{R}^d$ with $\|x_i\| \leq B$ and $\mathcal{H} = \{x \mapsto \langle w, x\rangle : \|w\| \leq 1\}$. Show

$$\mathbb{E} \sup_{\|w\| \leq 1} \frac{1}{n} \sum_i \varepsilon_i \langle w, x_i\rangle = \frac{1}{n} \mathbb{E} \Big\| \sum_i \varepsilon_i x_i \Big\| \leq \frac{1}{n} \Big(\sum_i \|x_i\|^2 \Big)^{1/2} \leq \frac{B}{\sqrt{n}},$$

and combine with Example 5.5 and Proposition 5.6 (quoting that composing with a 1-Lipschitz loss does not increase the Rademacher complexity) to obtain a generalization bound independent of d.

Exercise 5.16 (the Robbins–Monro noise)**.** type B Let $|\xi_k| \leq c$ and $\eta_k = \eta/k$. For $m \geq 2$ let $M_n = \sum_{k=m}^n \eta_k \xi_k$, $n \geq m$. (a) Show the steps are bounded by $\eta c/m$ and $\langle M \rangle_n \leq \eta^2 c^2/(m-1)$ for all n. (b) Deduce from Theorem 5.11

$$\mathbb{P}\Big(\exists\, n \geq m : \Big| \sum_{k=m}^n \eta_k \xi_k \Big| \geq u \Big) \leq 2 \exp\Big(- \frac{u^2}{2\big(\eta^2 c^2/(m-1) + \eta c u/(3m)\big)} \Big).$$

(c) Compare with Theorem 3.10(ii).

Exercise 5.17 (bounded differences, two examples)**.** type B (a) Let L_n be the length of the longest increasing subsequence of n i.i.d. uniform points in $[0,1]$. Show L_n has bounded differences 1, so $\mathbb{P}(|L_n - \mathbb{E}L_n| \geq u) \leq 2e^{-2u^2/n}$. (It is known that $\mathbb{E}L_n \sim 2\sqrt{n}$ and that the fluctuations are of order $n^{1/6}$.) (b) Let B_n be the minimal number of unit bins needed to pack n i.i.d. sizes in $[0,1]$. Show B_n has bounded differences 1, and that fluctuations of order $\sqrt{n}$ do occur (compare B_n with the sum of the sizes).

Exercise 5.18 (Freedman versus Azuma for SGD)**.** type C In Exercise 2.12 take $\eta_k = \eta/k$, $\eta > \frac{1}{2}$, and state-dependent noise $\varepsilon_{k+1} = s(\theta_k)\zeta_{k+1}$ with ζ_k i.i.d. signs and $s(\theta) = \min(|\theta|, 1)$. (a) Show that for fixed n the noise term $N_n = \sum_{k<n} w_{n,k} s(\theta_k)\zeta_{k+1}$ is the terminal value of a martingale with quadratic variation $\sum_{k<n} w_{n,k}^2 s(\theta_k)^2$. (b) Write the Azuma and Freedman bounds for $\mathbb{P}(|N_n| \geq u)$. (c) Show that on the event $\{|\theta_k| \leq \delta \text{ for } k \geq n/2\}$ Freedman's variance term is smaller than Azuma's by a factor of order δ^2, and set up a bootstrap argument for the rate of $\theta_n \to 0$. (d) Simulate and compare.

Exercise 5.19 (the operator norm)**.** type C Let M be an $N \times d$ matrix with independent entries. (a) If the entries lie in $[-1,1]$, show $\|M\|_{\mathrm{op}}$ has bounded differences 2 in each entry, so $\mathbb{P}(\|M\|_{\mathrm{op}} - \mathbb{E}\|M\|_{\mathrm{op}} \geq u) \leq e^{-u^2/(2Nd)}$. Compare the scale $\sqrt{Nd}$ with $\mathbb{E}\|M\|_{\mathrm{op}} \asymp \sqrt{N} + \sqrt{d}$ (Lectures 9–11), and check that grouping the entries into rows gives the same scale. (b) If the entries are i.i.d. $\mathcal{N}(0,1)$, show $M \mapsto \|M\|_{\mathrm{op}}$ is 1-Lipschitz for the Frobenius norm, and use the Gaussian concentration box with $\mathbb{E}\|M\|_{\mathrm{op}} \leq \sqrt{N} + \sqrt{d}$ (Lecture 11) to get $\mathbb{P}(\|M\|_{\mathrm{op}} \geq \sqrt{N} + \sqrt{d} + u) \leq e^{-u^2/2}$. (c) Which property of $\|\cdot\|_{\mathrm{op}}$ does (b) use that bounded differences cannot express? (Talagrand's convex distance inequality is the bounded-variables result that can.)